

# Shorter Die Battles (and Order Statistics)

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We rederive the main result in [1] in a shorter way, without combinatorics. If we have  $m$  independent and identically distributed (i.i.d.) random variables (r.v.'s)  $X_1, \dots, X_m$ , with  $X_i = X$  having cumulative distribution function (cdf)  $F_X(x) = \Pr(X \leq x)$ , then the r.v.  $X_{(m)} = \max_i X_i$  has cdf

$$F_{X_{(m)}}(x) = \Pr(X_{(m)} \leq x) = \prod_{i=1}^m \Pr(X_i \leq x) = (\Pr(X \leq x))^m = F_X^m(x).$$

Let  $X$  be a uniform r.v. with support  $\mathcal{X} = \{1, 2, \dots, k\}$  modelling a fair die with  $k$  faces. For  $x \in \mathcal{X}$  the cdf of  $X$  is

$$F_X(x) = \frac{x}{k}.$$

If  $Y_{(n)}$  is the maximum of the i.i.d. r.v.'s  $Y_1, \dots, Y_n$  with  $Y_i = X$  and independent of  $X_1, \dots, X_m$ , then

$$\begin{aligned} \Pr(X_{(m)} > Y_{(n)}) &= \sum_{a > b} \Pr(X_{(m)} = a, Y_{(n)} = b) \\ &= \sum_{a=2}^k \Pr(X_{(m)} = a) \Pr(Y_{(n)} < a) \\ &= \sum_{a=2}^k (F_{X_{(m)}}(a) - F_{X_{(m)}}(a-1)) F_{Y_{(n)}}(a-1) \\ &= \sum_{a=2}^k (F_X^m(a) - F_X^m(a-1)) F_X^n(a-1) \\ &= \frac{1}{k^{m+n}} \sum_{a=2}^k (a^m - (a-1)^m) (a-1)^n. \end{aligned}$$

## References

- [1] S.J. Miller. “Die Battles and Order Statistics”. Class Notes from Math 162: Statistics, Brown University, Spring 2006  
[https://web.williams.edu/Mathematics/sjmillier/public\\_html/BrownClasses/162/DieBattle.pdf](https://web.williams.edu/Mathematics/sjmillier/public_html/BrownClasses/162/DieBattle.pdf).

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[\[back\]](#)